

AS4545 - Undergraduate Research Project Report



Interaction of Expansion Waves: A Method of Characteristics Framework

Name: Purna Ananthkrishnan
Roll Number: AE22B003

Contents

1	Introduction	2
2	Theoretical Formulation	3
2.1	Self-Similar Solution of the Euler Equations	3
2.2	Characteristic Relations	3
2.3	Solution to the Ordinary Differential Equations	4
2.4	Determination of the Piston Velocity	5
2.5	Discretisation	5
2.6	Interaction	5
2.6.1	Wall - Wave Interaction	5
2.6.2	Wave - Wave Interaction	6
3	Results	7
3.1	Interaction of an Expansion Fan with a Wall	7
3.2	Interaction of Two Expansion Fans Moving in Opposite Directions	9
3.2.1	Case 1: Symmetric Interaction	9
3.2.2	Case 2: Asymmetric Interaction	10
3.3	Comparison of Symmetric Interaction to Reflection off a Wall	12
4	Conclusions	13

1 Introduction

The study of wave interactions in compressible flows has been a subject of considerable interest in gas dynamics since the early development of the field. The interactions of shock waves have been extensively studied, with foundational work establishing a comprehensive treatment of interactions between shock waves, rarefaction waves, and contact surfaces in one-dimensional flows.[1][2]The problem of shock wave interaction is well-posed and has been approached analytically, numerically and experimentally across a wide range of flow conditions. While shock wave interactions have continued to receive significant attention in the literature owing to their relevance in a wide variety of high-speed flows, the interaction of expansion waves with one another and with solid boundaries has received comparatively less attention.

The interaction of expansion waves is nonetheless of considerable physical importance. In shock tube flows, an expansion fan propagates into the driver section upon diaphragm rupture and subsequently reflects off the end wall, altering the thermodynamic state of the gas. Such interactions lead to significant reductions in pressure, density, and temperature in the affected region. More generally, the head-on interaction of oppositely travelling expansion waves can produce regions of low pressure and temperature from regions that were initially at high stagnation conditions. Despite this physical relevance, a systematic characterisation of such interactions in the one-dimensional unsteady setting remains limited in the existing literature, motivating the present study.

The analysis of the interaction of one-dimensional expansion waves in the inviscid limit can be theoretically approached using the method of characteristics.[2][3] Since expansion waves are themselves characteristics of the governing Euler equations, the method provides a natural and exact framework for their study. By exploiting the self-similar structure of the governing equations, the coupled partial differential equations can be reduced to ordinary differential equations, the solutions to which describe the variation of the thermodynamic state of the fluid across the expansion fan. In order to compute the changes in the state of the fluid due to individual wave interactions, the continuous fan is discretised into a finite number of characteristic waves, each carrying a small jump in the thermodynamic state, allowing the interactions (both with solid walls and with opposing expansion waves) to be computed systematically through isentropic and characteristic relations.[4][5]

The configuration considered here is based on a shock-tube-like setup, wherein the expansion fan is generated by a piston moving from rest to a steady velocity, determined by the pressure ratio across the driver and driven sections. The framework applies to both left- and right-moving expansion fans, with the solution for one following directly from the other by symmetry. The continuous fan is discretised into a finite number of characteristic waves, with the number of waves chosen to balance the ideal infinitesimal jumps across the isentropic waves with the associated computational cost.

In this work, the method of characteristics is applied to study the interaction of expansion waves in a one-dimensional shock-tube-like configuration. Three cases are considered: the reflection of a single expansion fan off a solid wall, the head-on interaction of two symmetric expansion fans, and the head-on interaction of two asymmetric expansion fans. The symmetric interaction is further compared to the wall reflection case to verify their equivalence by symmetry arguments. The remainder of this report is organised as follows: Section 2 presents the theoretical formulation, including the self-similar solution of the Euler equations, the characteristic relations, and the interaction algorithms. Section 3 discusses the results for each case, and Section 4 contains the concluding remarks.

2 Theoretical Formulation

The theoretical framework developed incorporates the solution of the inviscid Euler equations in a simple shock-tube-like configuration. The coupled partial differential equations are transformed into ordinary differential equations by introducing a similarity variable. The resulting ordinary differential equations can then be solved using characteristic relations. The solution of these ODEs provides the variation of the state variables across the regions of interest. These regions are discretised into a desired number of waves to balance the ideal infinitesimal jumps across the isentropic waves with the associated computational cost. Individual wave interactions, as well as wave-wall interactions, can then be computed using isentropic and characteristic relations.

2.1 Self-Similar Solution of the Euler Equations

The one-dimensional Euler equations are considered, with the energy equation reformulated in terms of pressure as shown below.

$$\begin{aligned}\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} &= 0, \\ \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} &= 0, \\ \frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} + \gamma p \frac{\partial u}{\partial x} &= 0.\end{aligned}\tag{1}$$

For simple compression and expansion waves, the characteristic wave speeds remain constant along the corresponding characteristic trajectories. This motivates the introduction of the similarity variable

$$\varepsilon = \frac{x}{t}\tag{2}$$

which allows the governing partial differential equations to be reduced to ordinary differential equations. The transformed equations are as follows

$$\begin{aligned}(\rho u)' - \varepsilon \rho' &= 0, \\ uu' + \frac{p'}{\rho} - \varepsilon u' &= 0, \\ up' + \gamma pu' - p' \varepsilon &= 0.\end{aligned}\tag{3}$$

Here, ' denotes differentiation with respect to the variable ε .

2.2 Characteristic Relations

An isentropic wave, of wavespeed c and the local speed of sound a , where $a = \sqrt{\gamma RT}$ with fluid velocity u obeys the following characteristic relation

$$c = a \pm u\tag{4}$$

Where the positive equation is applicable to right-running waves and the negative to left-running waves. Given that the entire process is isentropic, $P \propto \rho^\gamma$ is valid. Thus,

$$a^2 = \frac{dP}{d\rho} \propto \gamma \rho^{\gamma-1}\tag{5}$$

$$a \propto \rho^{\frac{\gamma-1}{2}} \Rightarrow \frac{da}{a} = \frac{\gamma-1}{2} \frac{d\rho}{\rho} \Rightarrow \frac{a}{\rho} d\rho = \frac{2}{\gamma-1} da\tag{6}$$

Along the characteristics,

$$du \pm \frac{a}{\rho} d\rho = 0 \quad \text{or} \quad du \pm \frac{2}{\gamma - 1} da = 0 \quad (7)$$

Integrating through the wave region,

$$u = \pm \frac{2}{\gamma - 1} (a - a_0) \quad (8)$$

Where the positive equation is applicable to right-running waves and the negative to left-running waves.¹

2.3 Solution to the Ordinary Differential Equations

Considering a left-moving expansion fan (as shown in figure 1), the left-most state (0) has the state variables p_0, ρ_0, a_0 with $u_0 = 0$. The right most state is defined by the velocity u_p (the piston velocity). As seen before, the similarity variable introduced, ε , is representative of the characteristic wave speeds. Here, for the i^{th} wave, the characteristic wave speed, $c_i = -dx/dt = -\varepsilon$, accounting for them moving towards the left.

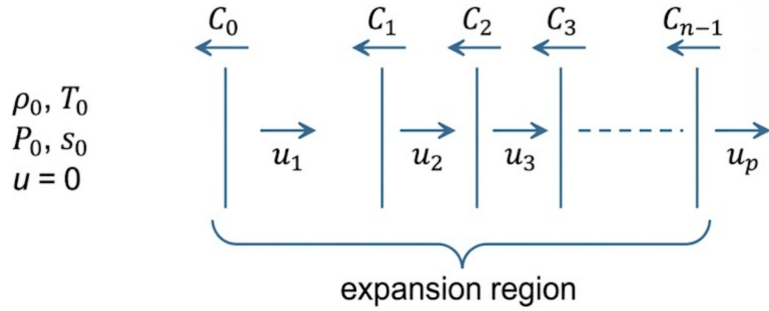


Figure 1: Schematic Diagram of the Characteristic Waves in a 1-D Left-Moving Expansion Fan

To simplify calculations, a new variable η is introduced,

$$\eta = \frac{\varepsilon}{a_0} \quad (9)$$

The ordinary differential equations can be reformulated in terms of η

$$\begin{aligned} \frac{d(\rho u)}{d\eta} - (\eta a_0) \frac{d\rho}{d\eta} &= 0, \\ u \frac{du}{d\eta} - \eta a_0 \frac{du}{d\eta} + \frac{1}{\rho} \frac{dP}{d\eta} &= 0, \\ u \frac{dP}{d\eta} - a_0 \eta \frac{dP}{d\eta} + \gamma P \frac{du}{d\eta} &= 0. \end{aligned} \quad (10)$$

Clearly, $\eta = -1$ corresponds to the left boundary of the expansion fan, thus the boundary conditions for the above set of ordinary differential equations are: $u(-1) = 0$, $P(-1) = P_0$ and $\rho(-1) = \rho_0$. The equations can now be solved using the characteristic relations to obtain u , P and ρ as functions of the variable η . The solutions are given in equation 11.

¹Here it has been taken that u also carries an appropriate sign (+ for moving right, -, for moving left). If u were to be just the magnitude, then for the piston moving left, the signs for the left and right running waves must be reversed; however, as it will be seen further, when having both left and right running expansion fans in one problem, it becomes necessary to incorporate the appropriate sign of the fluid velocity, thus the positive equation has been used for all right running fans and negative for left.

$$\begin{aligned}
u(\eta) &= \frac{2a_0(1+\eta)}{(\gamma+1)} \\
\rho(\eta) &= \rho_0 \left[\frac{(\gamma+1)}{2-\eta(\gamma-1)} \right]^{\frac{2}{\gamma-1}} \\
P(\eta) &= P_0 \left[\frac{2}{\gamma+1} - \left(\frac{\gamma-1}{\gamma+1} \right) \eta \right]^{\frac{2\gamma}{\gamma-1}}
\end{aligned} \tag{11}$$

The conditions on the right of such a left-moving expansion fan would be determined by the velocity u_p . In a shock tube, this would be the steady-state velocity of the piston, which is determined by the pressure difference across it.

2.4 Determination of the Piston Velocity

A similar procedure to the above can be carried out for a right-moving compression fan to obtain the state across the compression region in terms of a similarity variable. The right-most velocity obtained for the left-moving expansion fan is matched to the left-most velocity obtained for the right-moving compression fan, to obtain the velocity u_p in terms of the ratio of the left-most and right-most pressures of the expansion and compression fan, respectively. The velocity u_p is determined as

$$u_p = \frac{1}{k} \left(\frac{P-1}{P+1} \right) \tag{12}$$

Here, $k = (\gamma-1)/2a_0$ (assuming that both driver and driven sections have the same a_0) and $P = (P_{0e}/P_{0c})^{\frac{\gamma-1}{2\gamma}}$ with P_{0e} and P_{0c} the initial pressures of the driver and driven sections respectively.

2.5 Discretisation

The solution for the states in the expansion fan region given in section 2.3 is continuous starting from $\eta = 1$ and ending at η_p such that $u(\eta_p) = u_p$. In order to analyse the interaction of the individual expansion/characteristic waves, the region is discretised into a finite number of waves, n . The discretisation is carried out based on a uniform jump in η .

Given that the expansion fan is isentropic, the error in the discretisation is evaluated based on the change in entropy of the system. As shown in figure 1, the i^{th} characteristic wave has state i to its left and state $i+1$ to its right. The framework is equally applicable to right-moving expansion fans, since the solution follows directly by symmetry.

2.6 Interaction

Two possible interactions have been considered: the reflection of the wave with the wall and the interaction of two waves travelling in opposite directions. The interaction of two waves travelling in the same direction isn't possible in an expansion fan as the wave ahead will always have a higher wave velocity.

2.6.1 Wall - Wave Interaction

As a left-moving expansion wave reflects off a wall, the conditions to its right remain unchanged. The wall also imposes a zero-velocity condition. The state of the fluid near the wall and the wave speed are changed by this interaction, as shown in figure 2.

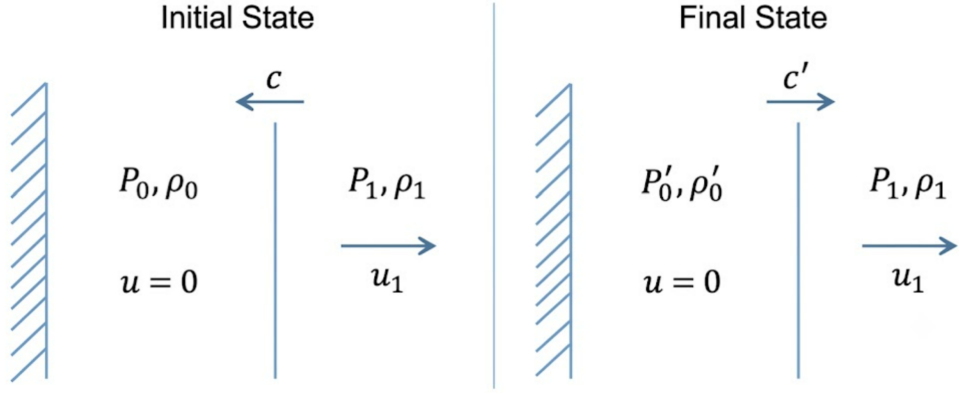


Figure 2: Schematic Diagram of a Left-Moving Expansion Wave Reflecting from a Wall

From 8, for the reflected wave

$$a'_0 = a_1 - \frac{\gamma - 1}{2} u_1$$

From a'_0 , the modified wave speed c' can be found, as $c' = a'_0$. The modified state P'_0 and ρ'_0 can also be found from a'_0

$$\rho'_0 = \rho_1 \frac{a'_0}{a_1}^{\frac{2}{\gamma-1}}$$

$$P'_0 = P_1 \frac{\rho'_0}{\rho_1}^\gamma$$

2.6.2 Wave - Wave Interaction

When two expansion waves travelling in opposite directions interact, they are expected to pass through each other without generating a contact discontinuity, since the flow remains isentropic. A schematic of this interaction is shown in Figure 3. The left and right states remain unchanged by the interaction. The wave speeds and the middle state are altered by the interaction.

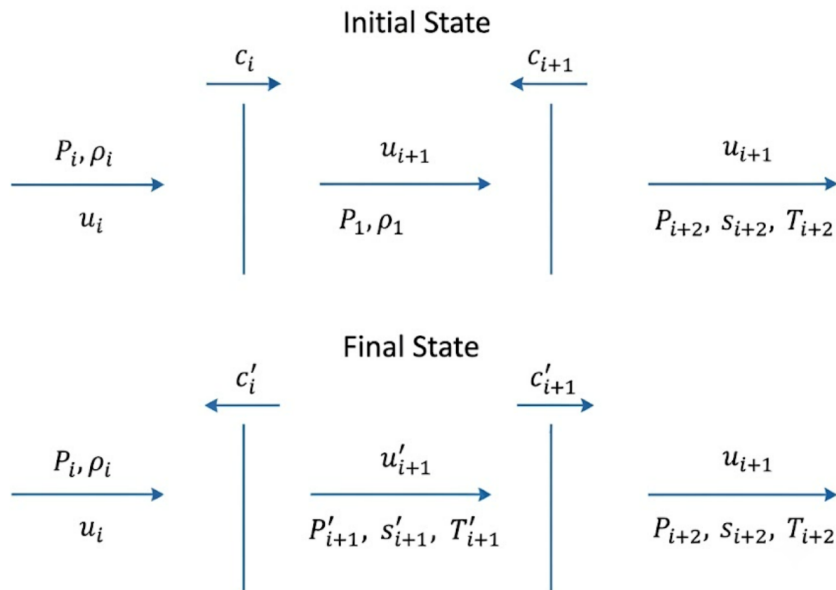


Figure 3: Schematic Diagram of the Interaction of Two Opposite Travelling Expansion Waves

The speed of sound for the modified $i + 1^{th}$ state can be found using either the i^{th} or the $i + 2^{th}$ state using equation 8. In order to reduce the error that may occur due to discretisation, the average of the two results is taken, as shown below

$$a'_{i+1} = \frac{1}{2} \left(a_i + a_{i+2} + \frac{\gamma - 1}{2} (u_i - u_{i+2}) \right)$$

The fluid velocity in the $i + 1^{th}$ state is

$$u'_{i+1} = \frac{2}{\gamma - 1} (a_{i+1} - a_{i+2}) + u_{i+2}$$

The new wave speeds (with appropriate signs) can be found in terms of a'_{i+1} and u'_{i+1} as

$$\begin{aligned} c'_i &= u'_{i+1} - a'_{i+1} \\ c'_{i+1} &= u'_{i+1} + a'_{i+1} \end{aligned}$$

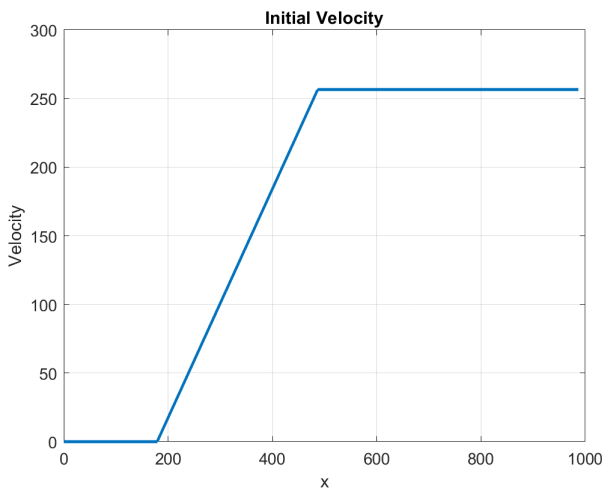
Finally, the $i + 1^{th}$ state can be found as

$$\begin{aligned} \rho'_{i+1} &= \rho_{i+2} \frac{a'_{i+1}}{a_{i+2}}^{\frac{2}{\gamma-1}} \\ P'_{i+1} &= P_{i+2} \frac{\rho'_{i+1}}{\rho_{i+2}}^{\gamma} \end{aligned}$$

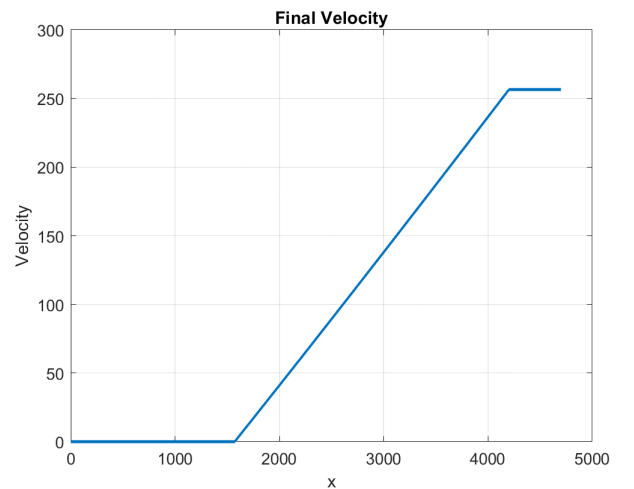
3 Results

3.1 Interaction of an Expansion Fan with a Wall

The system is set up with the wall located at $x = 0$. The pressure ratio is set as 5, and the initial temperature T_0 is set to 500 K, resulting in a steady-state piston velocity of approximately 257 m/s. The expansion fan region is discretised into 1000 characteristic waves, and the change in entropy of the system is found to be negligible, validating the isentropic assumption. The simulation was run for 6.92 seconds. The initial and final states are shown in the plots in figures 4 - 7.

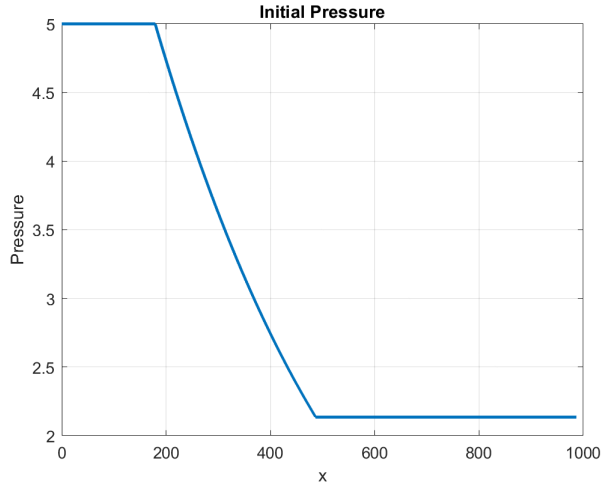


(a) Initial Variation of u with x ($t = 0$)

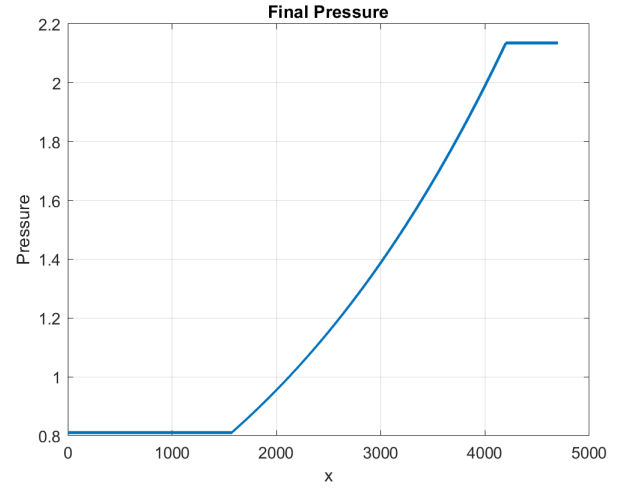


(b) Final Variation of u with x ($t = 6.92$)

Figure 4: Velocity Variation in Expansion Fan Region

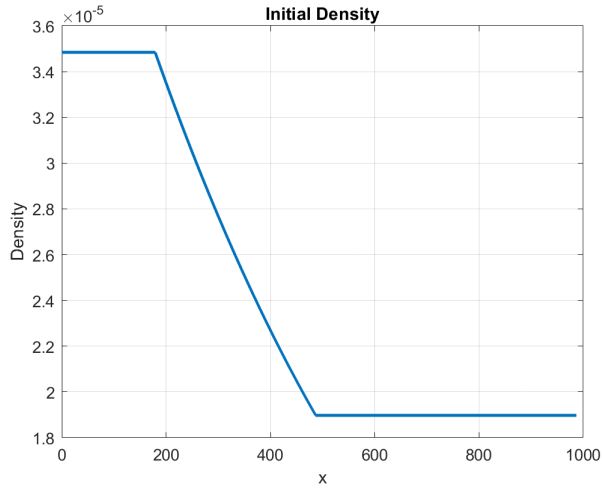


(a) Initial Variation of P with x ($t = 0$)

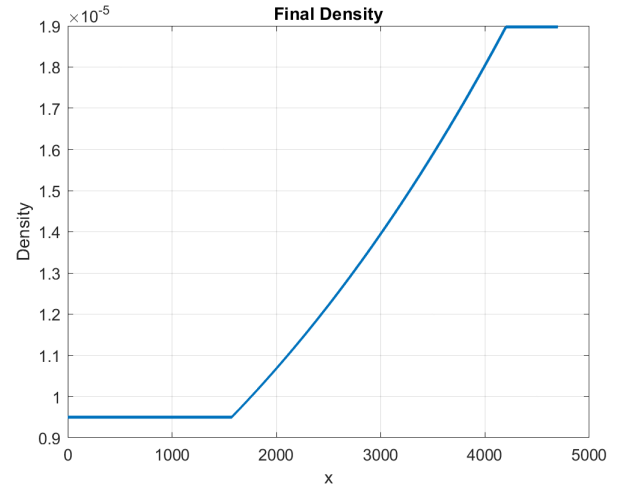


(b) Final Variation of P with x ($t = 6.92$)

Figure 5: Pressure Variation in Expansion Fan Region

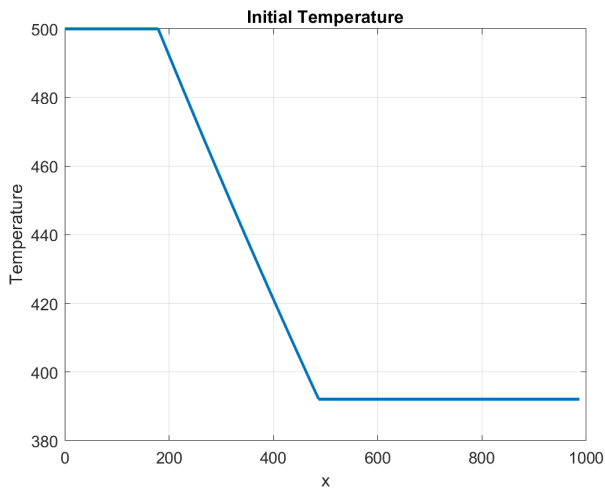


(a) Initial Variation of ρ with x ($t = 0$)

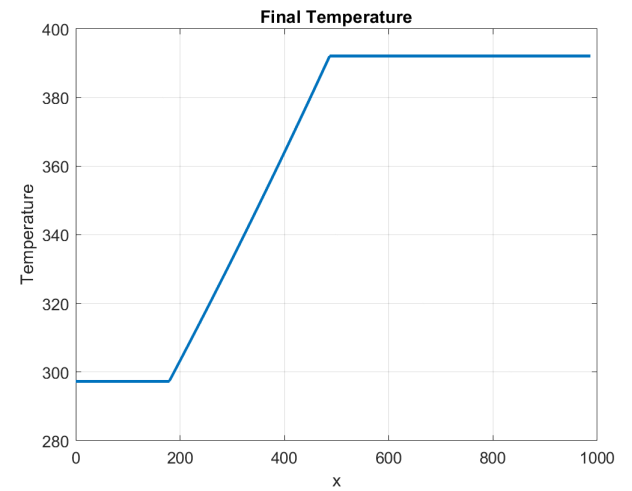


(b) Final Variation of ρ with x ($t = 6.92$)

Figure 6: Density Variation in Expansion Fan Region



(a) Initial Variation of T with x ($t = 0$)



(b) Final Variation of T with x ($t = 6.92$)

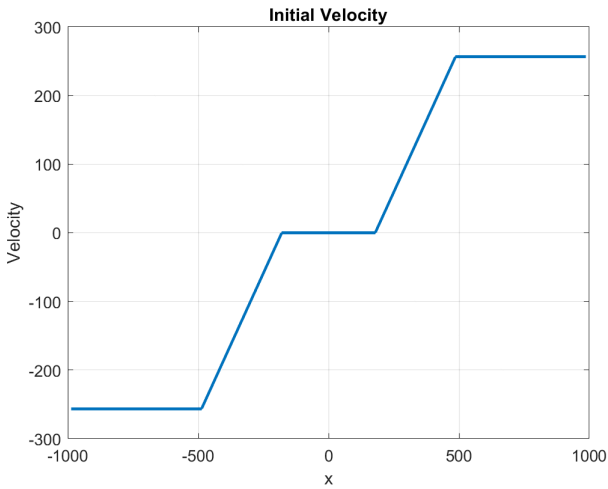
Figure 7: Temperature Variation in Expansion Fan Region

The results indicate that there is a significant reduction in the pressure, temperature and density near the wall due to the reflection of the expansion fan.

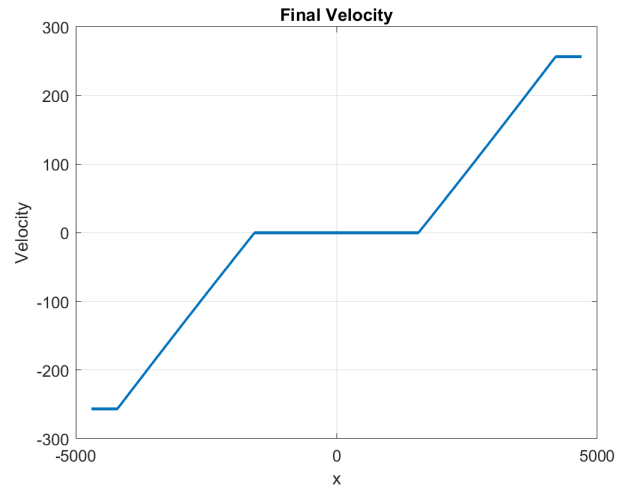
3.2 Interaction of Two Expansion Fans Moving in Opposite Directions

3.2.1 Case 1: Symmetric Interaction

Two symmetric expansion fans (one left-moving and one right-moving) were set up under identical conditions to those used in the wall interaction case, with a pressure ratio of 5 and $T_0 = 500$ K. Initially, both fans are symmetrically placed about $x = 0$. Each fan was discretised into 1000 characteristic waves, and the simulation was run for 6.92 seconds. The initial and final states are shown in the plots in figures 8 - 11.

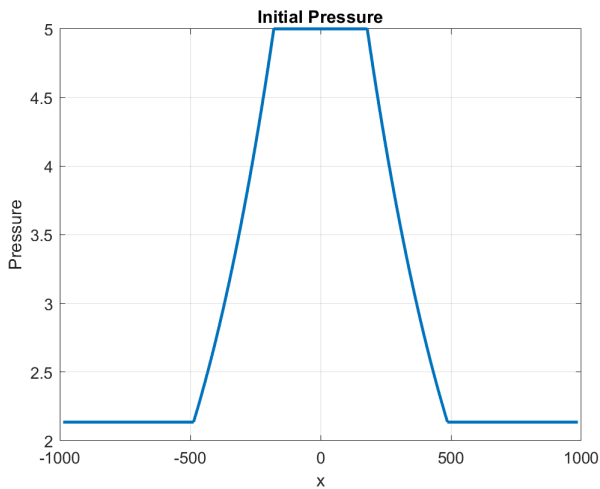


(a) Initial Variation of u with x ($t = 0$)

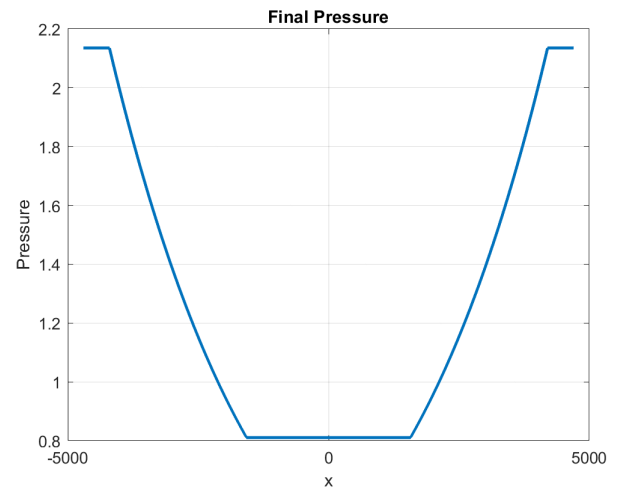


(b) Final Variation of u with x ($t = 6.92$)

Figure 8: Velocity Variation in Expansion Fan Region

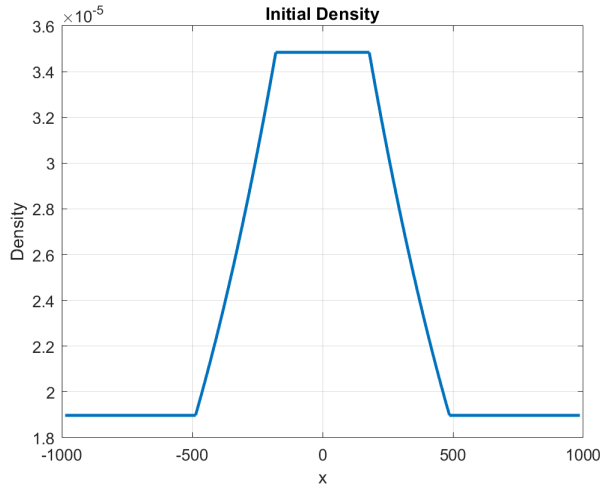


(a) Initial Variation of P with x ($t = 0$)

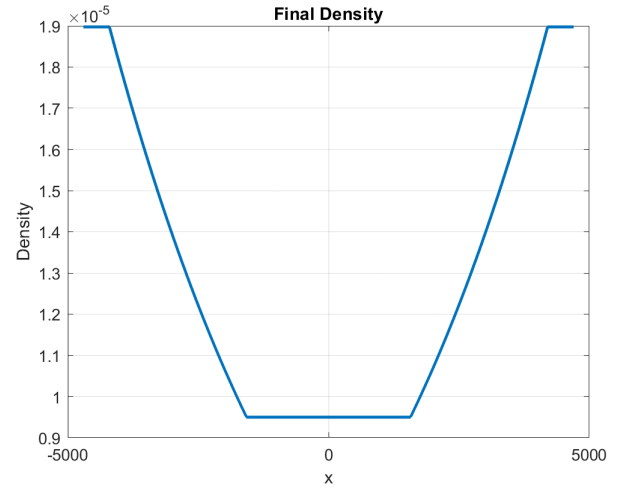


(b) Final Variation of P with x ($t = 6.92$)

Figure 9: Pressure Variation in Expansion Fan Region

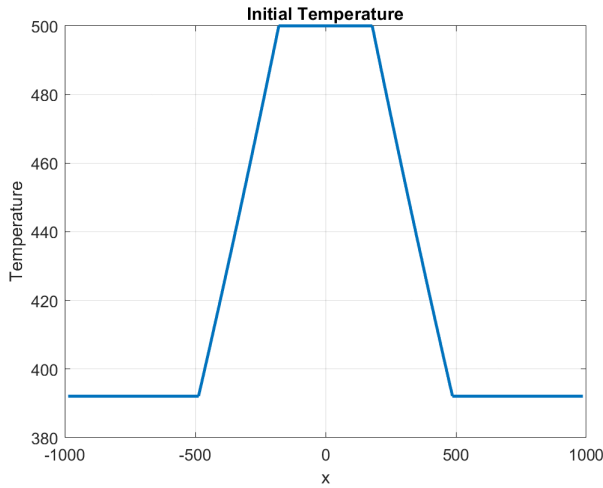


(a) Initial Variation of ρ with x ($t = 0$)

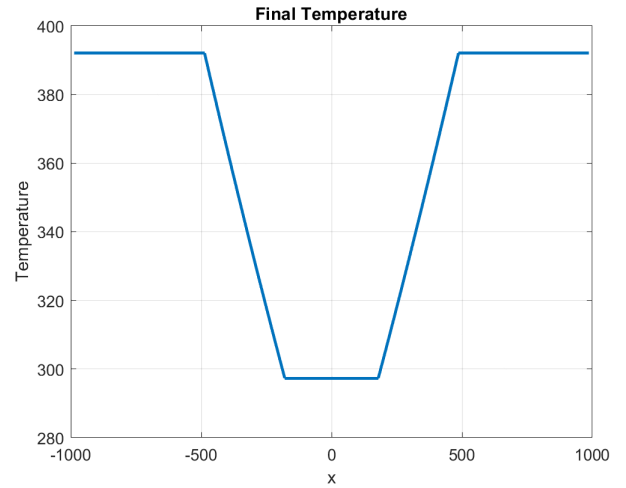


(b) Final Variation of ρ with x ($t = 6.92$)

Figure 10: Density Variation in Expansion Fan Region



(a) Initial Variation of T with x ($t = 0$)



(b) Final Variation of T with x ($t = 6.92$)

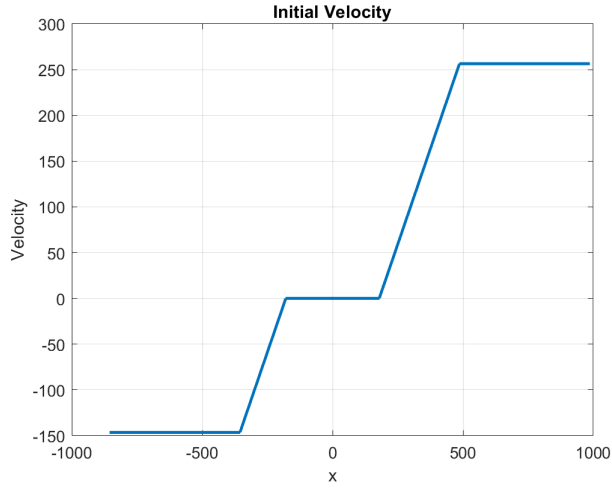
Figure 11: Temperature Variation in Expansion Fan Region

Similar to the wall reflection case, there is a significant drop in pressure, density, and temperature in the region between the expansion fans after they have completely interacted with one another. It is also interesting to note that even though no velocity condition has been imposed on the region between the two fans, it still remains zero after their interaction.

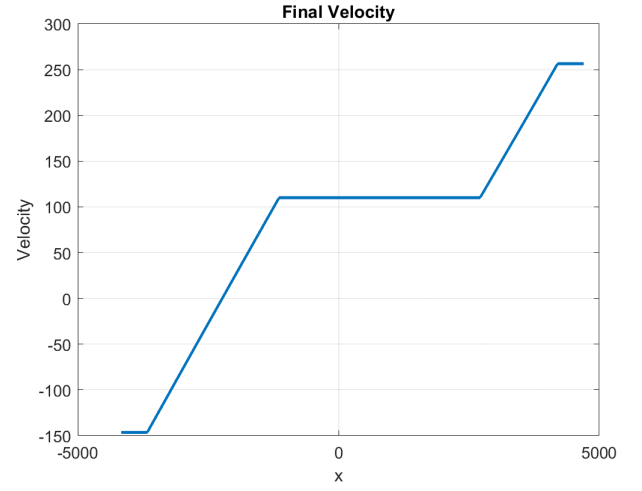
3.2.2 Case 2: Asymmetric Interaction

In order to study the effect of asymmetry on the interaction, the pressure ratio of the right-moving fan was reduced from 5 to 2.5 (half that of the left-moving fan) while all other conditions were kept identical to the symmetric case. This asymmetry in the initial conditions is expected to result in an asymmetric final state, with the two fans no longer being mirror images of each other after interaction. The initial and final states are shown in the plots in figures 12 - 15.

The results of the asymmetric interaction indicate that two oppositely moving expansion fans pass through each other rather than reflecting off one another. The drop in pressure, density, and temperature in the interaction region is not as high as in the symmetric case, owing to the lower pressure ratio of the right-moving fan. It is also noted that unlike the symmetric case, the velocity in the interaction region between the fans after they have passed through each other is no longer zero.

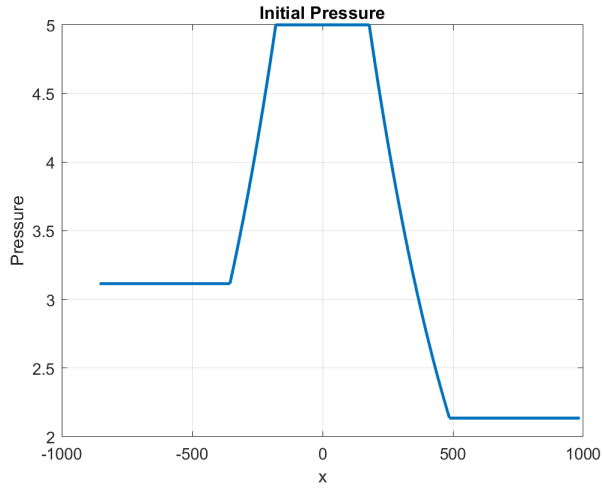


(a) Initial Variation of u with x ($t = 0$)

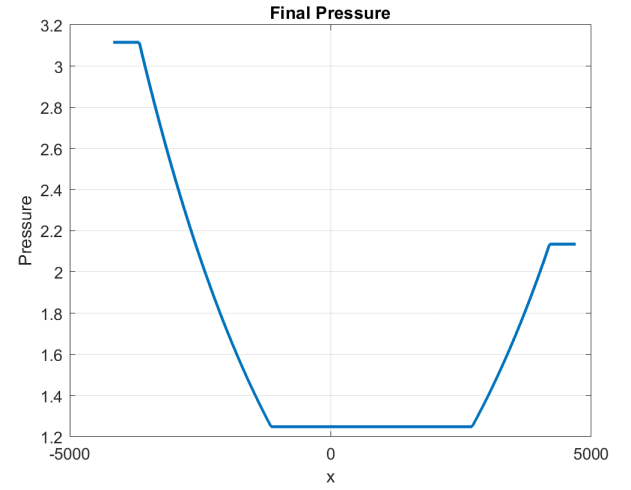


(b) Final Variation of u with x ($t = 6.92$)

Figure 12: Velocity Variation in Expansion Fan Region

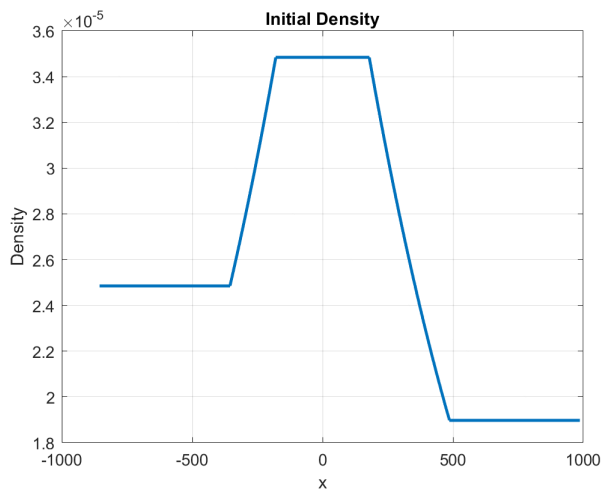


(a) Initial Variation of P with x ($t = 0$)

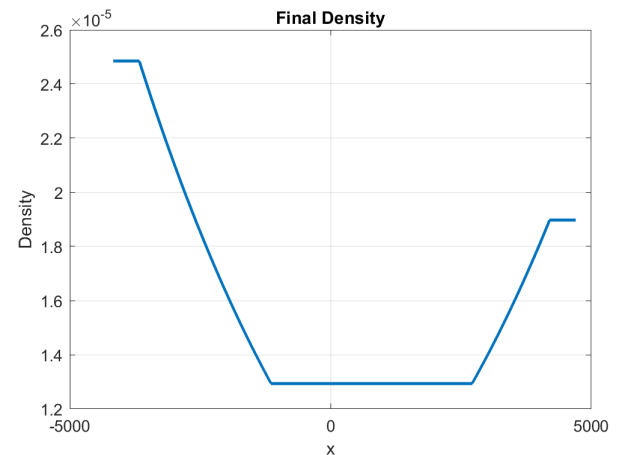


(b) Final Variation of P with x ($t = 6.92$)

Figure 13: Pressure Variation in Expansion Fan Region

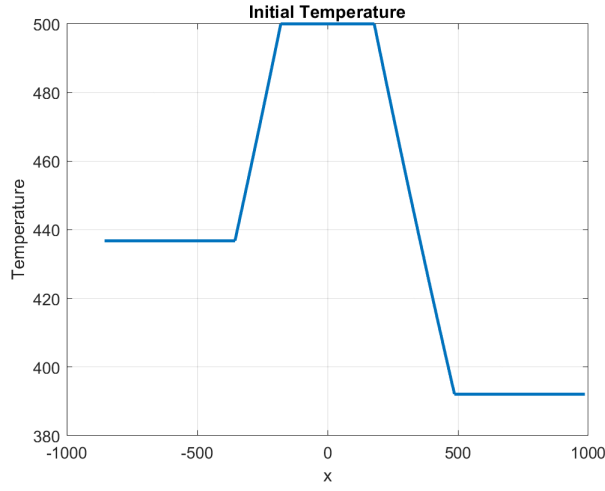


(a) Initial Variation of ρ with x ($t = 0$)

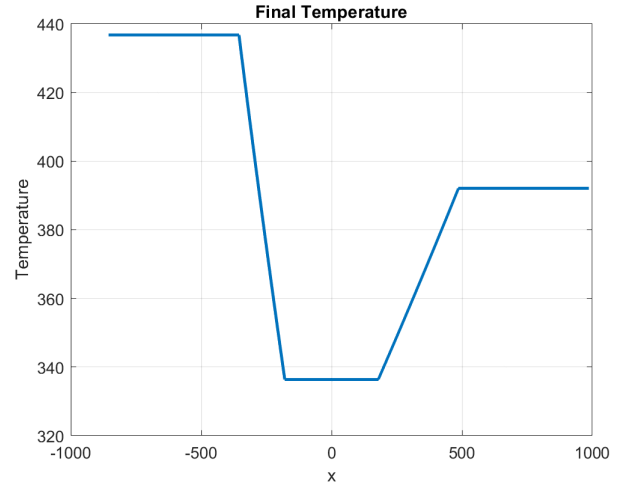


(b) Final Variation of ρ with x ($t = 6.92$)

Figure 14: Density Variation in Expansion Fan Region



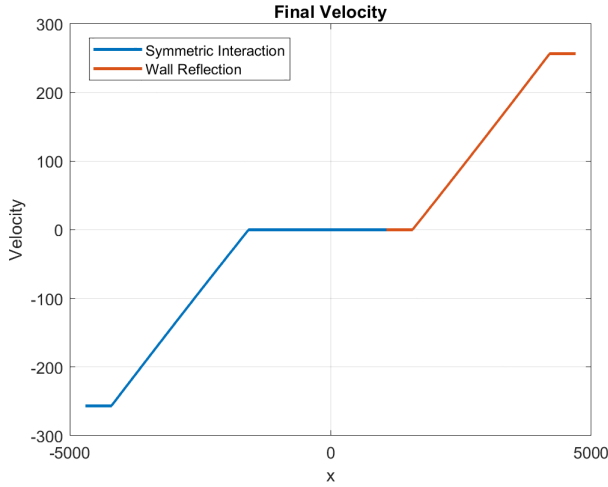
(a) Initial Variation of T with x ($t = 0$)



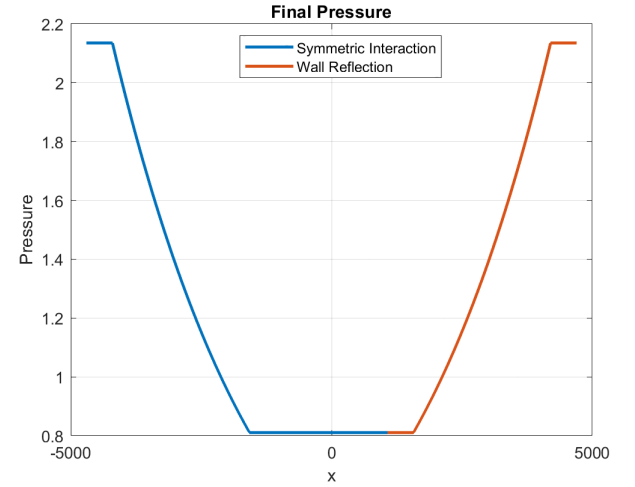
(b) Final Variation of T with x ($t = 6.92$)

Figure 15: Temperature Variation in Expansion Fan Region

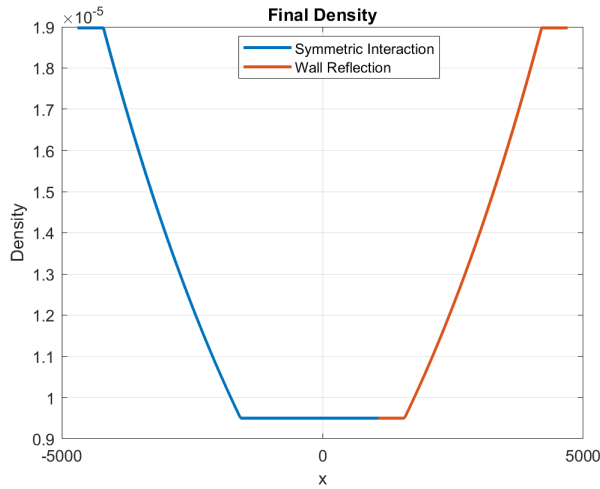
3.3 Comparison of Symmetric Interaction to Reflection off a Wall



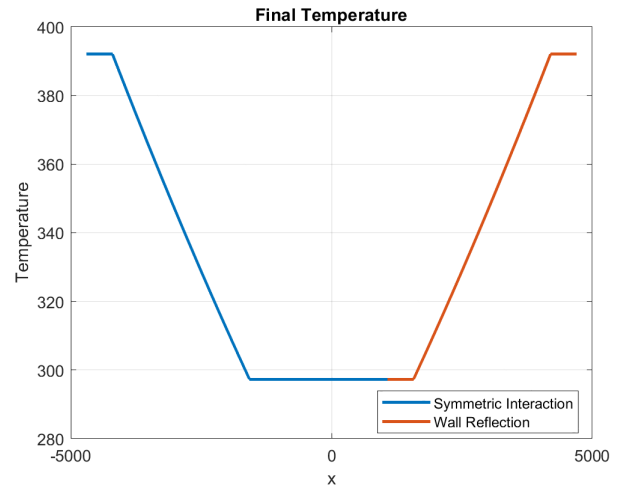
(a) Variation of u with x ($t = 6.92$)



(b) Variation of P with x ($t = 6.92$)



(c) Variation of ρ with x ($t = 6.92$)



(d) Variation of T with x ($t = 6.92$)

Figure 16: Comparison of the Final States of the Symmetric Interaction and the Wall Reflection Cases

To verify the equivalence between the symmetric wave interaction and the reflection of a single expansion fan off a wall, the final states of both cases are compared directly. By symmetry, the midpoint $x = 0$ in the symmetric interaction imposes a zero-velocity condition, which is identical to the boundary condition imposed by the wall in the wall reflection case. Therefore, the two problems are expected to yield identical results for $x > 0$. The final states of both cases are overlaid in the plots in Figure 16, confirming this equivalence.

4 Conclusions

A method of characteristics framework was used to study the interaction of expansion waves in a one-dimensional inviscid setting. Three cases were studied: the reflection of a single expansion fan off a solid wall, the head-on interaction of two symmetric expansion fans, and the head-on interaction of two asymmetric expansion fans. The wall reflection results showed a significant reduction in pressure, density, and temperature in the region near the wall following the interaction. The symmetric wave interaction produced qualitatively similar results and was shown to be equivalent to the wall reflection case, consistent with the argument that the zero-velocity condition at the midpoint mirrors that imposed by the wall. The asymmetric interaction confirmed that two oppositely moving expansion fans pass through each other, with the degree of reduction in the thermodynamic state governed by the relative strengths of the two fans. Unlike the symmetric case, the velocity in the interaction region was found to be non-zero, owing to the asymmetry in the initial conditions. Overall, the framework developed provides a systematic approach to study expansion wave interactions. The results obtained are consistent with the expected physical behaviour.

References

- [1] G. Ben-Dor, O. Igra, and T. Elperin. *Handbook of Shock Waves*. Academic Press, San Diego, 2001.
- [2] R. Courant and K.O. Friedrichs. *Supersonic Flow and Shock Waves*. Interscience Publishers, New York, 1948.
- [3] Harry Sauerwein. Numerical calculations of multidimensional and unsteady flows by the method of characteristics. *Journal of Computational Physics*, 1, 1967.
- [4] E.F. Toro. *Riemann Solvers and Numerical Methods for Fluid Dynamics*. Springer, Berlin, 3rd edition, 2009.
- [5] R.V.K. Chakravarthy, Vedanth Nair, T.M. Muruganandam, and Santanu Ghosh. Analytical and numerical study of normal shock response in a uniform duct. *Physics of Fluids*, 2018.